Apache Solr & Moodle Integration — Complete Guide

Test Document for Tika File Indexing Verification

---

1. Introduction to Apache Solr for Moodle

Apache Solr is a highly reliable, scalable and fault tolerant open-source search platform built on Apache Lucene. For Moodle LMS installations, Solr provides dramatically improved search performance compared to the built-in MySQL full-text search. This document serves as both a configuration reference and a test artifact for verifying that Solr's Apache Tika content extraction pipeline correctly indexes PDF files including their textual content, metadata, and embedded links.

2. Multi-Tenant Architecture

The Eledia Solr setup implements a multi-tenant model where each Moodle instance receives an isolated Solr user and dedicated cores. Security is enforced at two layers: (1) Solr BasicAuth with per-tenant credentials stored in tenants.env, and (2) Caddy reverse proxy with URL-level routing that restricts each tenant to their own core endpoints only. In SolrCloud mode, Solr's RuleBasedAuthorizationPlugin additionally enforces collection-level isolation via the collection field in permission rules.

3. Tika Content Extraction

Apache Tika is bundled with Solr as the extraction module (SOLR_MODULES=extraction). Moodle submits file content to the /update/extract endpoint which extracts text from PDF, DOCX, PPTX, ODT and other formats. The extracted content is indexed in the file_content field, enabling full-text search across all course materials including lecture slides, research papers, and student submissions.

4. Key Concepts

Inverted Index: Solr stores terms as keys mapping to document lists — enables O(1) lookup for any search term.

Schema Management: Moodle uses ManagedIndexSchemaFactory to dynamically register fields via the Schema API at /schema.

Faceted Search: Enable content filtering by course, area, author, and date using Solr facet.field parameters.

Highlight Snippets: The hl=true parameter returns context-highlighted excerpts around matching search terms.

Replication: Solr backup and restore via /replication endpoint — used by Moodle for disaster recovery.

PDF Indexing: This very document tests the complete pipeline: upload → Tika extraction → index → searchable content.

5. Reference Links

Official Solr documentation: https://solr.apache.org/guide/

Moodle Global Search documentation: https://docs.moodle.org/en/Global_search

Apache Tika documentation: https://tika.apache.org/

6. Test Keywords for Search Verification

The following terms must be findable via Solr full-text search after this document is indexed through the Tika extraction pipeline: ELEDIA_TIKA_TEST_MARKER solr moodle apache lucene indexing extraction tika multi-tenant security basicauth faceting highlighting schema replication backup searchable content course material lecture slides research paper.
